Statement of Purpose

PhD Application in Economics (Latest Update: December 2025)

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1 Background Introduction

My name is Yongheng Hu from Zhejiang International Studies University. I will graduate in the summer of 2026 with a Bachelor degree of Economics. My undergraduate major was International Economics and Trade. However, the fields that truly interest me and motivate my application for doctoral studies are macroeconomics, behavioral economics (experimental economics) and political economy. And I intend to pursue a Ph.D. in Economics and dedicate myself to becoming a world leading scholar in the field of macroeconomics. My primary research interests will focus on macroeconomics, and the main objectives that drive my research are:

(**A**) *To analytically and empirically investigate why optimal theory in macroeconomics often fails to explain many problems in the real world?*

(**B**) *How can we comprehensively identify genuine and reliable heterogeneity in macroeconomics to improve traditional optimal theory?*

Regarding these two dimensions, I have preliminarily completed some fundamental works and papers during my undergraduate studies:

I have independently authored seven working papers in the field of macroeconomics. Four of these have been submitted to arXiv (Hu, 2025a; Hu, 2025b; Hu, 2025c; Hu, 2025d), one is in the process of final revisions and will be submitted to *American Economic Review: Insights* soon, and two are working in progress. Meanwhile, I have collaborated on the writing of six papers in the field of economics. Three of these have been published (Du et al., 2023; Xiong et al., 2025; Du et al., 2025), one is in the final stages of peer review and will be published soon, one is working in progress, and one received highly positive feedback and constructive suggestions after undergoing peer review by *Econometrica*. This paper is currently being revised and we plan to resubmit it to *Econometrica* in the future.

I have led two university students' research projects, both centered on issues in the field of macroeconomics. As a research assistant, I participated in four national and provincial economics research projects led by professors in total. During my tenure as a research associate, my working papers were accepted four times at economics academic conferences hosted by Chinese top universities, where I served as a presenter each time. Additionally, my individual research was accepted twice at the International Congress of Basic Sciences (*ICBS,2024*; *ICBS,2025*), where I presented my academic research posters.

2 Research Experience

I acquired foundational knowledge of economics through general university courses, but this constitutes only 10% of my current knowledge base. Most of my expertise stems from my research assistant work, specifically the process of collaborating with professors to draft, submit, revise and review academic papers. Of course, the main method is still self-learning under the comprehensive guidance of my professors. Based on my research interests and needs, I persist in studying advanced knowledge points and frontiers in the field of economics which I find compelling.

My initial motivation for applying to the PhD program in Economics stemmed from my involvement in research projects as a research assistant and my participation as a primary collaborator on a paper submitted to the *Econometrica*.

Professor Longzheng Du was my academic advisor at Zhejiang International Studies University from 2022 to 2024. Since my enrollment in 2022, I have served as a research assistant, deeply involved in Professor Du's National Natural Science Foundation project and writing multiple working papers. Under his guidance, I systematically studied fundamental economic theories and academic paper writing. After completing numerous data analysis tasks, I learned and strengthened my skills in economic experiments, specifically econometrics. Therefore, I have frequently observed in the empirical analysis sections of numerous working papers that many real situations are not consistent with optimal economic theory. For example, when conducting the heterogeneity analysis in my empirical research, I often found that the regression results for different subjects were inconsistent with the overall regression results, with some even failing to reach statistical significance. This implies that even if the overall regression results are consistent with optimal theory (some time even the overall regression results violate optimal theory), when we conduct group regression to preliminarily identify heterogeneity, the regression results for each group may violate optimal theory or be insignificant.

In June 2024, I was introduced to Dr. Mingyi Yang of Washington State University through the recommendation of Professor Longzheng Du. As a collaborator, I participated in Dr. Yang's academic paper writing by undertaking portions of the mathematical analysis, numerical simulation and empirical research. Then, I systematically engaged with and learned about the frontier areas of macroeconomic research with Dr. Yang's guidance. Meanwhile, I deeply understood that my current knowledge of economics and mathematics was insufficient to conduct truly valuable macroeconomic research. Unfortunately, constrained by my university's course selection policies and teaching resources, I was unable to study more advanced mathematics and economics at my undergraduate university, hence, I decided to study the specialized knowledge I needed for my research on my own with Dr. Yang's comprehensive guidance and teaching. I have now gained a preliminary understanding of methodologies including real analysis, functional analysis, measure theory, dynamic recursive methods, applied stochastic processes, behavioral economics and macroeconomic analysis.

I have also acquired foundational knowledge of heterogeneous agent models (Huggett, 1993; Aiyagari, 1994; Krusell and Smith, 1998; Adrian and Shin, 2010; Brunnermeier and Sannikov, 2014; Moll, 2014; Gabaix et al., 2016; Kaplan et al., 2018; Ahn et al., 2018; Gosselin et al., 2020; Bilal, 2023; Fernández-Villaverde et al., 2023; Maxted et al., 2025; Le Grand and Ragot, 2025) and Mean Field Games (MFG) (Achdou and Laurière, 2020; Achdou et al., 2022; Moll and Ryzhik, 2025) in macroeconomics through reading and studying relevant literature.

Crucially, I have preliminarily learned a powerful method from many insightful discussions with Dr. Yang, this new method may find broad application in solving problems in macroeconomics that traditional approaches struggle to solve effectively, namely, Path Integrals. In short: The optimal path is deterministic in classical framework of analysis. When stochastic processes are introduced, the optimal path becomes uncertain. Furthermore, after considering the heterogeneity, we can analyze all possible stochastic processes and calculate the average path through the path integrals. As shown in Figure 1:

The difference between the classic path $\bar{x}(t)$ and some possible alternative path $x(t)$ is the function $y(t)$. Since the paths must both reach the same end points, $y(t_a) = y(t_b) = 0$. In between these end points $y(t)$ can take any form. Since the classic path is completed fixed, any variation in the alternative path $x(t)$ is equivalent to the associated variation in the difference $y(t)$. Thus, in a path integral, the

path differential $\mathcal{D}x(t)$ can be replaced by $\mathcal{D}y(t)$, and the path $x(t)$ by $\bar{x}(t) + y(t)$. In this form $\bar{x}(t)$ is a constant for the integration over paths. Furthermore, the new path variable $y(t)$ is restricted to take the value 0 at both end points. This substitution leads to a path integral independent of end-point positions.

Here, the heterogeneity comes from agents' capabilities or endowments rather than behavioral consequences. As shown in Figure 1, under the conventional view, if two types of agents transition from point a to point b and simultaneously remain at point b, they might be classified as homogeneous. What I wish to emphasize is that even when the consequences are identical, if the transition paths (processes) are different, they should still be recognized as heterogeneous agents.

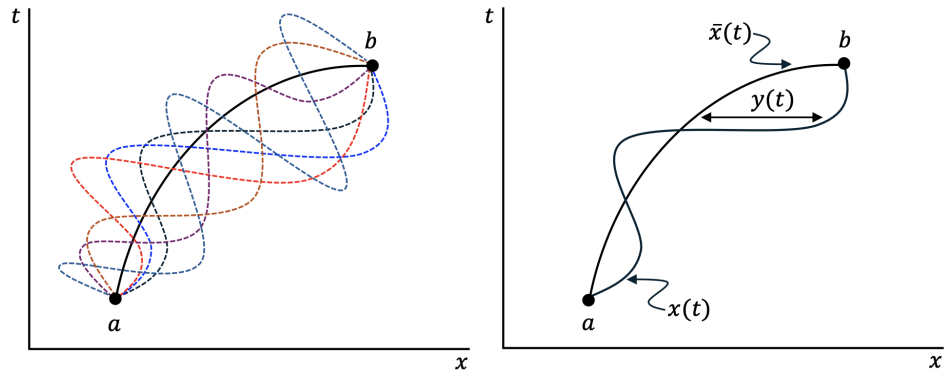

Figure 1: Path and all Paths

Conventional analysis focuses on identifying heterogeneity through economic consequences. However, I contend that relying solely on consequences to identify heterogeneity while ignoring the economic pathways is weak. True heterogeneity should stem from the paths determined by agents' resource endowments, i.e., Resource (Endowment) $\rightarrow$ Path (State) $\rightarrow$ Heterogeneity (Distribution) $\rightarrow$ Economics (Equilibrium). This implies that even if economic consequences are identical, i.e., they stay at the same level of equilibrium or partial equilibrium, differing pathways to achieve them should still be recognized as evidence of heterogeneous agents. Because the stability of equilibrium is endogenous to the agents' pathways.

A typical example: Two countries simultaneously adopt the same development pattern. Decades later, they may reach the same economic level, which seems to suggest homogeneity. However, the changes in national capacity differences during this transition are dynamic. This implies that each country has multiple actual paths under the same development pattern. Analyzing which specific path was taken historically is merely an average probability. Moreover, differing development paths can affect the stability of their equivalent economic levels. Therefore, considering both the paths to reach results and the stability of those results, these two countries should be regarded as heterogeneous. Consequently, identifying all possible paths and measuring the stability of their corresponding results becomes a critical task.

Hence, during this year of studying and collaborating with Dr. Yang, I came to understand a crucial point: The essence of macroeconomics is a distribution. That is, we must rely on the distribution of microeconomic variables to study macroeconomic issues rather than solely on aggregate shocks. A classic example is the distribution of income and wealth.

When we study macroeconomics at the perspective of "*Distribution*", heterogeneity becomes continuous and infinite. I thus believe that under such an analytical framework, the theory for identifying heterogeneity will differ from traditional econometrics or macroeconomics: Conventionally, heterogeneity only stems from economic outcomes, and we typically classify heterogeneous agents based on the results of their economic behavior. In this context, heterogeneity is static and discrete. Based on MFG and related theories, I contend that we should also identify heterogeneity from the agents themselves. That

is, the heterogeneity of heterogeneous agents does not only stem from the outcomes of their economic decisions but is also influenced by certain resource endowments inherent to them (such as learning or investment abilities, cognitive resources, etc.). Since resource endowments can be varied by exogenous or endogenous factors, the heterogeneity of agents in this context is dynamic and continuous. Regarding this perspective, I have identified two classic theories in economics that could be improved: Prospect Theory and the Lucas Critique. In the following sections of my research statement, I will outline how I attempt to "*Update Prospect Theory*" and "*Challenge the Lucas Critique*" based on my current research.

3 Main Research Statement

3.1 Heterogeneity in State

The first article is "*Heterogeneity of the Nonrival Data Model*", which is my writing sample for PhD application and will be submitted to *American Economic Review: Insights* soon. This paper took nearly three years to complete, starting with the initial representative agent model (Hu, 2025a), progressing to the identification of heterogeneity (Hu, 2025c) and finally being completed in September 2025.

In this paper, I extend the traditional analytical framework of nonrival data (Jones and Tonetti, 2020) and introduce the heterogeneous agents model. I systematically analyze the social planner model and general equilibrium model with nonrival data , as well as the optimal allocation of nonrival data by heterogeneous agents. When I analyze optimal allocation theory again based on empirical results, I find that actual situations violate the optimal theory: According to the optimal allocation theory, firms with high ability converge to zero at a faster speed than those with low ability in allocating nonrival data. However, in reality, especially in a capital environment with reduced financial frictions, firms with high ability will continuously allocate nonrival data to establish monopolistic dominance.

We do not consider the state of heterogeneous agents under traditional optimal theory, hence, nonrival data dissipation is constant, whereas in practice, it depends on the state of the firm, approximating a quadratic function. According to this contradiction, I introduced the agent's "state" as a shock and constructed a State-Dependent KFE (SD-KEF) to attempt to resolve some problems of the theory. When I introduced the state-dependent shock, traditional economic or mathematical methods become ineffective, such as the Laplace Transform (Gabaix et al., 2016) and Eigenvalue-Eigenfunction Decomposition (Alvarez and Lippi, 2022). These methods cannot obtain analytical solutions, leading to obstacles in further analysis. I attempted to use a new type of method, namely "Path Integral" (Feynman, 1948), to derive the analytical solution for the SD-KFE.

It is worth noting that I wish to emphasize how path integral extends far beyond the applications discussed in this paper when considering macroeconomic issues. Most notably, we can employ it to solve problems related to wealth distribution. For instance: Firms' wealth exist in three states, high, medium and low, they dissipate wealth according to their states. After obtaining the analytical equilibrium of wealth distribution by using MFG, we can define the state function of firm wealth as a quadratic form and substitute it into the traditional Time-Dependent KFE (TD-KFE). This implies that the dissipation of firm wealth depends on the firm's state rather than a constant dissipation rate. We can then employ the path integral to investigate the impact of state shocks under partial equilibrium conditions analytically. Further applications will be detailed in the paper that we plan to submit to *Econometrica* in the future.

Therefore, when we can express an economic variable x_t as a Brownian Motion (BM), such as wealth, income, taxation, exchange rates or interest rates, etc, we can transform the constant dissipation β into a certain form of dissipation function $\beta(x)$ based on the heterogeneity of agents' states (government, firms

or consumers). Then, we can perhaps solve our problem analytically by using the path integral. That is:

$$\text{BM}: dx_t = \mu dt + \Sigma d\mathcal{W}_t$$
$$\text{TD}-\text{KFE}: \frac{\partial p(x,t)}{\partial t} = -\mu\frac{\partial p(x,t)}{\partial x} + \frac{1}{2}\Sigma^2\frac{\partial^2 p(x,t)}{\partial x^2} - \beta p(x,t) + \beta\delta(x)$$
$$\text{SD}-\text{KFE}: \frac{\partial p(x,t)}{\partial t} = -\mu\frac{\partial p(x,t)}{\partial x} + \frac{1}{2}\Sigma^2\frac{\partial^2 p(x,t)}{\partial x^2} - \beta(x)p(x,t) + \beta(x)\delta(x)$$

3.2 New Assumption to Decision-making

The second article is "*How Big Data Dilutes Cognitive Resources, Interferes with Rational Decision-making and Affects Wealth Distribution?*" (Hu, 2025b). This paper was accepted for presentation at the undergraduate academic forum hosted by the Institute of New Structural Economics at Peking University and received an Outstanding Paper Award. Because of this paper, I was fortunate to meet Professor Shenzhe Jiang of Peking University, who offered me tremendous encouragement to continue refining this paper and related ideas.

One of the key developments in macroeconomics over the past few decades has been the introduction of heterogeneity and the construction of macro models incorporating individual differences. However, the rational expectations assumption in current macroeconomic models with heterogeneous agents is not only computationally infeasible but also conceptually inconsistent with reality. We could have three principles for revision: (**1**) Computational feasibility. (**2**) Empirical consistency. (**3**) The endogeneity of beliefs to the model's reality (Moll, 2025). This paper primarily explores issues related to the third principle. And I propose the following three propositions:

(**A**) *Rational state is determined by rational resources, namely, cognitive resources* R_t.

(**B**) *Rational state and decisions cannot be observed simultaneously.*

This is quite natural. In reality, if agents could observe their own rational state while making decisions, their decisions should be rational at all times. However, this is not the fact. Hence, we can assume that the agent's decision $\widehat{x}_t^n \in \mathbb{R}^n$ and its rational state $\widehat{y}_{R_t}^m \in \mathbb{R}^m$ are in a commutation relationship, that is, they cannot be observed simultaneously. Measured by $\Delta \in \mathbb{R}^+$, we obtain $\left[\widehat{x}_t^n, \widehat{y}_{R_t}^m\right] = \widehat{x}_t^n\widehat{y}_{R_t}^m - \widehat{y}_{R_t}^m\widehat{x}_t^n = \Delta$. Thus, the analytical solution of Δ could define rational decisions.

(**C**) *Rational state is asymmetric.*

Asymmetry signifies that rational state will shift when rational resources are affected, rather than the conventional assumption that only one rational state exists for a particular agent. For example, traditional prospect theory employs a single curve to describe decisions (Kahneman and Tversky, 1979). This implies that the degree of deviation from rational decisions varies for the same decision under different rational states. We assume the probability density distribution of an agent's decisions under any rational state is $P(\widehat{x}_{t_i}^n|\widehat{y}_{R_{t_i}}^m)$. If the agent can simultaneously observe the rational state while making decision, the decision degenerates to $\widehat{x}^n$, i.e., $\Delta = 0$. Then: $\mathbb{E}[P(\widehat{x}_{t_0}^n|\widehat{y}_{R_{t_1}}^m)] - \mathbb{E}[P(\widehat{x}^n)] \neq \mathbb{E}[P(\widehat{x}_{t_0}^n|\widehat{y}_{R_{t_2}}^m)] - \mathbb{E}[P(\widehat{x}^n)]$.

The core idea of these three propositions is that agents should possess "rational resources". We need to analyze the issue of rational or irrational expectations based on the endowment of rational resources in agents, rather than only judging whether agents are rational or irrational based on the results of their economic decisions. This proposition may currently be applied in the following two directions:

(**1**) **The Prospect Theory**: The first concerns updating Prospect Theory. Conventionally, agents' decision adjustments and environmental uncertainty are modeled as a static curve (Kahneman and Tversky, 1979), because in this analytical framework we judge agents' rationality based on economic consequences. I contend that the determination of an agent's rationality should concentrate more on the agent's

own resource endowment. These resources, named cognitive resources, determine whether an agent can make rational decision adjustments. Since cognitive resources, like other resources in reality, exist in a distributed state, their distribution changes when subjected to endogenous or exogenous shocks. Consequently, the relationship between agents' decision adjustments and environmental uncertainty becomes endogenous to the agents' cognitive resources, forming a dynamic distributed curve.

In short, the traditional description of prospect theory curves is static, with only one curve. When we incorporate agents' cognitive resources, two distinct thresholds emerge: One representing decisions made when agents possess maximum cognitive resources and face minimal environmental uncertainty, and another reflecting decisions made when agents possess minimal cognitive resources and face maximum environmental uncertainty. This scenario demonstrates that the actual curve describing prospect theory should dynamically shift continuously between these two thresholds. As shown in Figure 2:

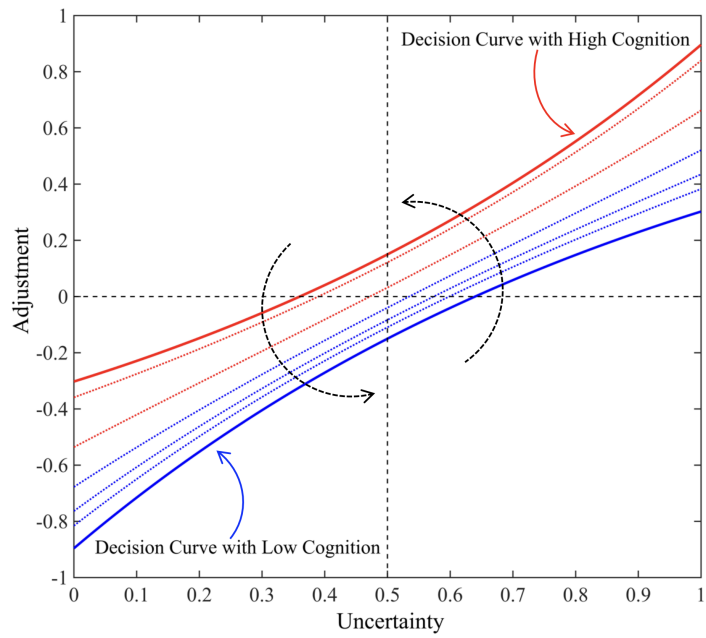

Figure 2: The Dynamics of Decision-making Curve

(**2**) **The Lucas Critique**: The second idea is an attempt to "challenge" the Lucas Critique. Conventionally, the Lucas Critique primarily describes how policies are formulated based on historical data, but agents adjust their behavior in response to these policies, potentially rendering the policies ineffective. However, when products are disseminated or sold in the form of data, they acquire a concept of "glossiness" that influences agents' perceptions of the products and disrupts their consumption decisions (Acemoglu et al., 2025). Applying this perspective to policy, when policies are disseminated through the media of data, they can influence agents' cognitive resources. This enables agents to allocate their cognitive resources toward the domains or directions desired by the policy, thus aligning their decisions with policy objectives. Therefore, policies can be effective under certain conditions. When agents' rationality can be significantly endogenous to the policy, their decisions represent not only "individual rationality" but also "policy rationality." In such cases, the Lucas Critique is weak and can be further refined.

Hence, at the conclusion of my working paper, I propose a hypothesis to refine (or challenge) the Lucas Critique. That is: policy can be effective under certain conditions. Effective policy should encompass two dimensions: "*Master Policy*" for planning economic objectives and "*Auxiliary Policy*" for allocating agents' rational resources. In short, if policy can possess stronger "directionality" rather than relying solely on quantitative targets, it can be effective. This perhaps make parts of the Lucas Critique invalid.

As this paper is still far from publication, I require further study and knowledge to enhance its coherence according to another two principles.

4 Summary

Since the 1980s, the core framework of macroeconomics has been dominated by DSGE (*Dynamic Stochastic General Equilibrium*) models. While these models are concise and efficient in analyzing macroeconomic fluctuations around a stable equilibrium point, they still have limitations.

From the 2008 global financial crisis to the current rise of artificial intelligence, unstable financial frictions, rapidly worsening wealth inequality, the development of public intelligence technologies and nonrival resources have made it difficult for real economics to be adequately explained by the representative agents. Therefore, we need to identify "*Endogenous Heterogeneity*" rather than solely relying on "*Exogenous Heterogeneity*". This involves applying "*Occam's Razor*" to construct heterogeneous agents model from the first principles, then studying issues such as wealth distribution inequality, social mobility, rational decision-making and how different groups respond to macroeconomic shocks.

Based on the work and papers I have completed so far, I have the following two insights:

(**A**) *The heterogeneity of agents should be simultaneously determined by the consequences of economic decisions and the agents' own resource endowments which is related to decision-making, rather than by any single dimension alone.*

(**B**) *If we can identify agents' genuine and complete heterogeneity and then incorporate it into economic models analytically, it may enhance the compatibility between optimal economic theory and reality.*

I wish to apply for the PhD program in Economics to receive more rigorous and professional training in the field, and to seek like-minded collaborators to work together on solving key problems that challenge the frontiers of economics.

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